

Year 10 Science
Energy and Change - Road Science
Safety Features of Cars

You have studied and researched the safety features found in cars designed for today's road users. Fantastic as it may seem, you have been appointed to the design team for Fiori Cars.

Your brief is to design a car suitable for use on Australian roads. It must be able to carry 5 people comfortably, and must be able to work in city and country environments. Off-road travel is **NOT** an option.

Safety is your prime concern.

Listed below are some of the major safety features found in cars today.

- Anti-lock Braking System (ABS).
- Air bags - driver, passenger, side curtain.
- Crumple zones.
- Inertia reel seat belts.
- Collapsible steering column.

Task

Using your knowledge gained in this topic, describe **ONE** safety design you would have in your car.

The following points must be addressed.

- How does the safety feature work? (Focus on its mechanics.)
- Explain the Physics principles behind the feature. (Describe the Science.)
- Why did you choose this feature? Why did you feel it was so important to include it?

Include simple and clearly labelled diagrams to help your research.

Presentation

You may present your research in any of the following forms.

- Booklet - A4 sheets stapled together.
- Poster - may have more than one sheet.
- Keynote presentation - recorded on a CD, USB thumb drive, or similar device.
- Podcast

Include a **detailed bibliography** listing all sources of information.

e.g. Cook, A.J. 2001, *Safety Features of Modern Cars*, Allen and Unwin, Melbourne, pp.216-224.

Einstein, A. 1905, *How Physics Can Be Applied to the Model T Ford*,
located at <http://howaboutthat.com>, accessed on 30 February, 2006.

